

Inner Queries & Views

Two last TYPES of queries, we do in basic SQL are Inner Queries and Views.

- We use **INNER QUERIES** when we want to compare an AGGREGATE value against the ROWS. Without subqueries you can compare to groups, **not specific rows**. Or if we want to SIMPLIFY a query.
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→ Like this

- Which Albums made more than average (**Subquery**) VS Show average musician salary (**Without**)

Which Albums made more than average?

```
SELECT albumtitle
FROM Album
WHERE sales > (SELECT AVG(sales) FROM Album)
```

WHERE, normally can't use **AGGREGATE**.

We can do ANYTHING we normally do in a normal query, inside that smaller one. Like an **INNER JOIN**, or a **GROUP BY**; So even if I don't want an aggregate check I can combine it here.

```
SELECT albumtitle
FROM Album
WHERE producerid IN (
    SELECT p.producerid
    FROM Producer p
    INNER JOIN Album a ON p.producerid = a.producerid
    WHERE a.sales > 1000000
);
```

Views, are a pretty simple thing. It's a NEW table that only has access to certain columns so "query people" can't see everything.

So, we CREATE a view by just doing a normal query, and saving that as a table.

> This returns HIGH SALARY ARTISTS.

```
SELECT a.artistname, SUM(p.fee) AS total_fees
FROM Artist a
INNER JOIN PERFORMANCES p ON a.artistid = p.artistid
GROUP BY a.artistid, a.artistname
HAVING SUM(p.fee) > (SELECT AVG(fee) FROM PERFORMANCES)
```

To make this a view, we just simply add these above.

```
GO

CREATE VIEW HighEarningArtists AS
```

Now, we can QUERY IT AS NORMAL.

```
SELECT *
FROM HighEarningArtists
```

This, is it for the SQL lab. If you want to continue.

Learn SELF JOINS, EXISTS, LIMIT.